

# Yixin Chen

PhD candidate, Peking University

Google Scholar   GitHub   ORCID

yixinchen0320@gmail.com

2311110791@stu.pku.edu.cn

yixinchen-ai.github.io

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## RESEARCH PROFILE

I work on medical image segmentation and learning across PET, CT and MRI. My clinical research focuses on cancer, fatty liver disease and multi-organ metabolism.

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## EDUCATION

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| <b>Peking University</b><br>PhD in Medical Technology                                           | September 2023-present |
| <b>King's College London</b><br>MSc in Artificial Intelligence, with Distinction                | 2019-2020              |
| <b>Xiamen University</b><br>BSc in Automation; second bachelor's degree in Mathematical Finance | 2015-2019              |

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## CHALLENGE RESULTS AND RESEARCH FUNDING

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>autoPET V: second place in Category 1</b><br>MICCAI autoPET V challenge, interactive PET/CT lesion segmentation.                                                | 2026                                        |
| <b>CMR-MULTI: second place</b><br>Multi-sequence cardiac MRI segmentation. Method paper selected for a CMRSeg challenge oral presentation.                         | 2026                                        |
| <b>REG<sup>2</sup>: fourth place</b><br>Method paper selected for a MICCAI 2026 challenge workshop poster.                                                         | 2026                                        |
| <b>NSFC doctoral research project</b><br>Principal investigator of a project on extracting and analysing multi-organ metabolic information from total-body PET/CT. | 2026-2027, NSFC grant for doctoral students |

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## PROFESSIONAL EXPERIENCE

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| <b>AirDoc</b><br><b>Algorithm Engineer</b><br>Developed a model to predict the gap between an implanted ICL lens and the natural lens after surgery, using eye measurements and lens size.<br>Worked on ultra-widefield fundus image analysis, clinical requirements and medical-device registration.                                                                                 | July-September 2022     |
| <b>BioMind</b><br><b>Medical Imaging Algorithm Engineer</b><br>Developed cardiac ultrasound models for image quality assessment, Doppler segmentation and measurement, septal defect detection and valve regurgitation analysis.<br>Worked with clinicians to integrate models into an ultrasound-assisted diagnosis system. Also contributed to an ovarian-tumour ultrasound system. | November 2020-June 2022 |

2026

**Whole-Body [<sup>18</sup>F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD**

EJNMMI, 2026  
Accepted

**Yixin Chen** (co-first author), Chuanzhi Zhou, Lijuan Guo, et al.

**NeuronCtrl: Geometry-Aware Safe Closed-Loop Generative Control for Neuronal Microenvironment Dynamics**

ICML, 2026  
Spotlight

Haowei Xu, **Yixin Chen**, W. Fu, Hongbin Han, Zhaoheng Xie  
Conference   OpenReview

**Neural Network Approximation Aided Kinetic Parameter Inference of Compartment Models in Dynamic PET**

IEEE TRPMS, 2026

Wenrui Shao, **Yixin Chen**, Ourui Fu, Tian Li, Zhi Yang, Nan Li, Hongbin Han, Xiangxi Meng, Zhaoheng Xie  
Paper

**PHATE-Net: Differentiable Pseudotime Learning for Trustworthy Disease Trajectories in PET**

CVPR Findings, 2026

**Yixin Chen**, Yan Wang, Wenrui Shao, Zhaoheng Xie  
Paper

2025

**Modality-projection universal model for comprehensive full-body medical imaging segmentation**

Nature Communications,  
2025

**Yixin Chen** (co-first author), Lin Gao, Yajuan Gao, Rui Wang, Jingge Lian, Xiangxi Meng, Yanhua Duan, Leiying Chai, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie  
Paper   arXiv   Code

**VP-SFDA: Visual Prompt Source-Free Domain Adaptation for Cross-Modal Medical Image**

Health Data Science,  
2025

**Yixin Chen**, Yan Wang, Zhaoheng Xie  
Paper

**Neural network-aided unsupervised input function estimation for dual-time-window PET Patlak analysis**

EJNMMI Physics, 2025

Wenrui Shao, Yarong Zhang, Fen Du, Fangxiao Cheng, **Yixin Chen**, Xiangxi Meng, Ying Liang, Zhaoheng Xie  
Paper

**A Physiology-Supervised Neural Operator Framework for Kinetic Parameter Estimation in Dynamic PET**

IEEE NSS/MIC/RTSD,  
2025

W. Shao, **Yixin Chen**, Z. Wang, R. Xu, X. Meng, Z. Xie  
Paper

2024

IEEE TMI, 2024

**PCNet: Prior Category Network for CT Universal Segmentation Model**

**Yixin Chen**, Yajuan Gao, Lei Zhu, Wenrui Shao, Yanye Lu, Hongbin Han, Zhaoheng Xie

Paper Code

MICCAI, 2024

Poster

**LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation**

**Yixin Chen**, Xiangxi Meng, Yan Wang, Shuang Zeng, Xi Liu, Zhaoheng Xie

Paper Code

IEEE TRPMS, 2024

**Structure-Enhanced Unsupervised Domain Adaptation for CT Whole-Brain Segmentation**

**Yixin Chen** (co-first author), Yajun Gao, Lei Zhu, Jianan Li, Yan Wang, Jiakui Hu, Hongbin Han, Yanye Lu, Zhaoheng Xie

Paper Code

IEEE NSS/MIC/RTSD,

2024

**Neural Network-Assisted Dual-time-window Acquisition: Modeling Incomplete Input Functions for Patlak Analysis**

W. Shao, Y. Zhang, **Yixin Chen**, X. Meng, Y. Liang, Z. Xie

Paper

2023

IEEE TMI, 2023

**FVP: Fourier Visual Prompting for Source-Free Unsupervised Domain Adaptation of Medical Image Segmentation**

Yan Wang, Jian Cheng, **Yixin Chen**, Shuai Shao, Lanyun Zhu, Zhenzhou Wu, Tao Liu, Haogang Zhu

Paper

IEEE EMBC, 2023

**Rethinking Disentanglement in Unsupervised Domain Adaptation for Medical Image Segmentation**

Yan Wang, **Yixin Chen** (co-first author), Yingying Zhang, Haogang Zhu

Paper

Frontiers in

Cardiovascular

Medicine, 2023

**Echocardiography-based AI for detection and quantification of atrial septal defect**

Xixiang Lin, Feifei Yang, **Yixin Chen** (co-first author), et al.

Paper

2022

IEEE EMBC, 2022

**MSGAN: Multi-Stage Generative Adversarial Networks for Cross-Modality Domain Adaptation**

Yan Wang, **Yixin Chen** (co-first author), Wenjun Wang, Haogang Zhu

Paper

Cancers, 2022

**A deep learning model system for diagnosis and management of adnexal masses**

Jingbo Li, **Yixin Chen** (co-first author), Minyu Zhang, et al.

Paper

Xixiang Lin, Feifei Yang, **Yixin Chen**, et al.  
Paper

## Preprints

arXiv, 2026

### PreSight: Preoperative Outcome Prediction for Parkinson's Disease via Region-Prior Morphometry and Patient-Specific Weighting

Yand Wang, Chen Zhang, Lanyun Zhu, **Yixin Chen**, Qunbo Wang, Yutong Bai, Jurgen Germann, Yinghong Wen, Shuai Shao  
arXiv

arXiv, 2023

### SL: Stable learning in source-free domain adaption for medical image segmentation

**Yixin Chen**, Yan Wang  
arXiv

## Challenge papers and conference abstracts

### Diagnosis-Conditioned Retrieval for Structured Pathology Workflows

MICCAI REG<sup>2</sup> challenge  
workshop, 2026  
Poster

**Yixin Chen**  
Paper

### Acquisition-Structured nnU-Net Ensembles for Multi-Sequence Cardiac MRI Segmentation and Quantification

MICCAI CMRSeg  
workshop, 2026  
Selected for oral  
presentation

**Yixin Chen**, Yizhen Peng  
Paper

### Failure-Mode-Driven Multimodal Mitral Valve Segmentation for the MVAA 2026 Challenge

MICCAI MVAA  
challenge workshop,  
2026  
Poster

**Yixin Chen**  
Paper

### Multiorgan Metabolic Dysregulation in Alzheimer's Disease: Total-body PET/CT Reveals Systemic Network Interactions with Amyloid Burden

EANM, 2025  
Oral abstract, OP-308

R. Xu, Y. Duan, L. Gao, **Yixin Chen**, H. Han, Z. Cheng, Z. Xie  
Programme

## PATENTS, TEACHING AND ACADEMIC SERVICE

### 12 invention patents

4 granted; 8 under substantive examination.

### Nature Communications

Early Career Co-Reviewer

### Computer-vision teaching

Recorded introductory and intermediate courses on machine learning and computer vision.

### Deep Learning Theory and Practice: PyTorch Case Studies

2021

Sole author. Published by Tsinghua University Press in 2021.